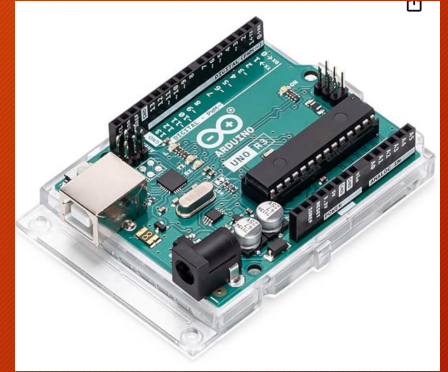
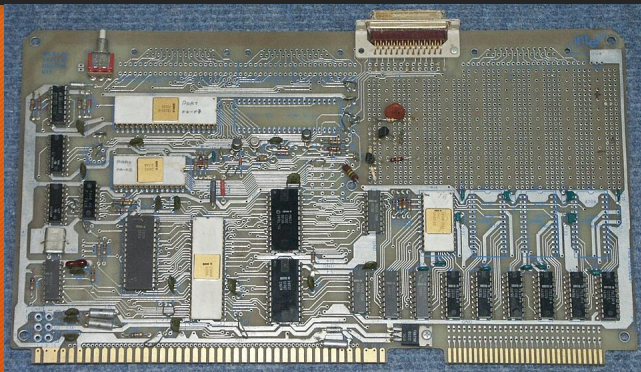


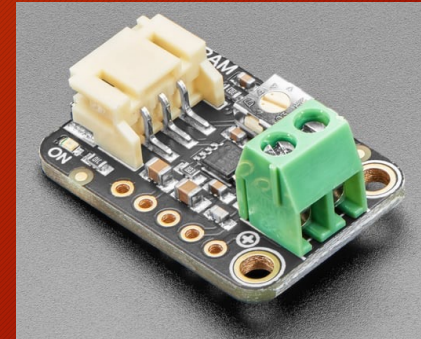
# CDL Presents Basic Electronics



1



Joe Wilkes, PhD  
WA2SFF



June 2025



# Week 4

2

Digital ICs

June 2025



# Digital ICs

3

- Logic ICs
- Field Programmable Gate Array (FPGA)
- Microprocessors
- Digital Signal Processor (DSP)
- Class D Audio Amplifier



# Logic ICs

4

- The earliest ICs implemented logic circuits that were previously implemented in complex Vacuum Tube or transistor circuits
- The first generation used Resistor-Transistor Logic: RTL
- Then Transistor-Transistor Logic: TTL
  - Low power TTL was an improvement ( L in part #)
  - Shottsky TTL improved the speed ( S in part #)
  - Low power Shottsky combined the two (LS in part #)
- Later designs used CMOS transistors and were called CMOS
  - Speeds were fast, up to several Hundred Megahertz
- Understanding this technology is important for restoring and rebuilding early computers and video games.
  - For example the Apple II used all off the shelf logic ICs
- A complex design might used hundreds of logic ICs
- Today logic ICs have been replaced with Field Programmable Gate Arrays (FPGA)

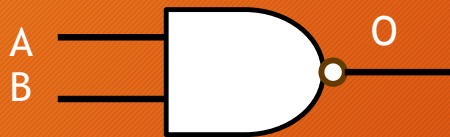
June 2025



# NAND Gate

5

NAND Gate



| A | B | O |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 1 | 0 |

The standard two-, three-, four- and eight-input NAND gates are available:

## CMOS

- 4011: Quad two-input NAND gate
- 4023: Triple three-input NAND gate
- 4012: Dual four-input NAND gate
- 4068: Mono eight-input NAND gate

## TTL

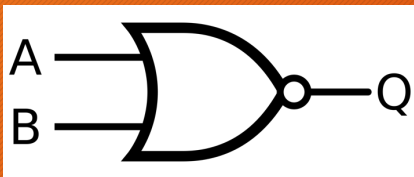
- 7400: Quad two-input NAND gate
- 7410: Triple three-input NAND gate
- 7420: Dual four-input NAND gate
- 7430: Mono eight-input NAND gate



# NOR Gate

6

NOR Gate



| A | B | Q |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 1 | 0 |

In the popular CMOS and TTL logic families, NOR gates with up to 8 inputs are available:

## CMOS

- 4001: Quad 2-input NOR gate
- 4025: Triple 3-input NOR gate
- 4002: Dual 4-input NOR gate
- 4078: Single 8-input NOR gate

## TTL

- 7402: Quad 2-input NOR gate
- 7427: Triple 3-input NOR gate
- 7425: Dual 4-input NOR gate (with strobe, obsolete)
- 74260: Dual 5-Input NOR Gate
- 744078: Single 8-input NOR Gate



# Flip Flop - 1

7

- Flip Flops are latches that have two stable states
- Each flip flop can implement 1 bit
- The output of a flip flop can input the next flip flop, thus forming a counter
- A row of flip flops can thus store binary numbers
- There are multiple types of flip flops, each with a different set of inputs and outputs, thus performing slightly different operations

[https://en.wikipedia.org/wiki/Flip-flop\\_\(electronics\)](https://en.wikipedia.org/wiki/Flip-flop_(electronics))

June 2025



# Flip Flop - 2

8

- Flip-flops and latches can be divided into common types:
  - SR ("set-reset"),
  - D ("data"),
  - T ("toggle"), and
  - JK
- Each type has a different set of logic to change its state
- Flip flops can be edge triggered or level triggered to change state
- The set-reset and the JK flip flop are the most common
- For more information see the Wikipedia page
  - [https://en.wikipedia.org/wiki/Flip-flop\\_\(electronics\)](https://en.wikipedia.org/wiki/Flip-flop_(electronics))



# Flip Flop - 3

9

- TTL
  - 7473: A standard J-K flip-flop
  - 7474: A D-type flip-flop
  - 74hc273: An octal D-type flip-flop
  - 74LS107: A J-K flip-flop
  - 74LS109: A J-K flip-flop with an inverted K input.
  - 74LS279: An R-S flip-flop
  - 74LS175: A quad D-type flip-flop with clear.
  - 74LS273: An octal D-type flip-flop with clear.
- CMOS
  - CD4013B: A dual D-type flip-flop
  - CD40174B: A hex D-type flip-flop with master reset.



# Field Programmable Gate Array

10

- A field-programmable gate array (FPGA) is a configurable integrated circuit that can be repeatedly programmed after manufacturing.
- FPGAs consist of an array of programmable logic blocks with a connecting grid, that can be configured "in the field" to interconnect with other logic blocks to perform various digital functions.
- FPGAs are used in
  - low production runs of custom-made products, and
  - in research and development,
  - where the higher cost of individual FPGAs is not as important, and cost and timeline of producing a custom part is not feasible.
- Applications for FPGAs include the telecommunications, automotive, aerospace, and industrial sectors
- FPGAs are often used to off load high speed processing that is difficult to perform in a CPU, GPU or DSP



# FPGA Programming

11

- FPGA are designed in a program that describes the blocks and interconnection of the blocks
- Code can be written, compiled, loaded, and tested very rapidly
- Custom parts can have turn around times measured in months and rework can also take months
- Once the code is written, turn around time can be measured in days or weeks
- VHDL and Verilog are the dominant languages for FPGA programming
- Different organizations tend to use one or the other
- Once one programming environment is learned, it is relatively easy to switch to the other one
- The software includes many libraries to reduce development time
  - A complete microprocessor is one common library
- New and faster and larger FPGAs are constantly being introduced
- It is best to do a survey of FPGAs and development boards when you start a new project
- Any list today would be obsolete very shortly

June 2025



# Microprocessors

12

- A microprocessor is a computer processor for which all the functions of a computer are on a single or small number of chips
- The first microprocessor was the Intel 4004
- It was followed up by the Intel 8008 and 8080
- Motorola developed the 6500, 6800 and later 68000 microprocessors
- In the 1980s, the cost was in the hundreds of dollars for the basic chip
  - support chips added even more cost
- TI, Signetics, RCA, AT&T, and others quickly developed their own designs
- Variations of all these designs were also made
- While it would be rare to use any of these in modern designs, other than emulation of older computers, anyone refurbishing or restoring old computer and video games would need to be familiar with them.

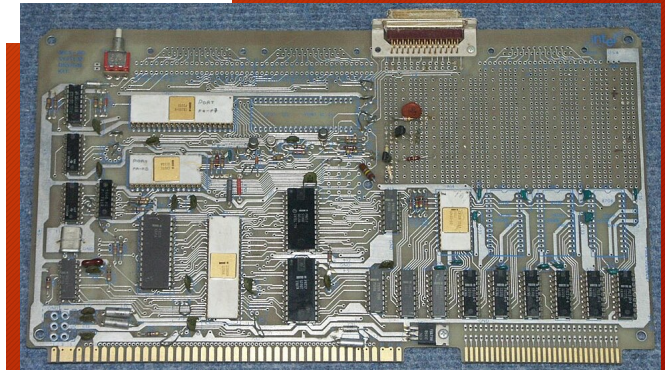
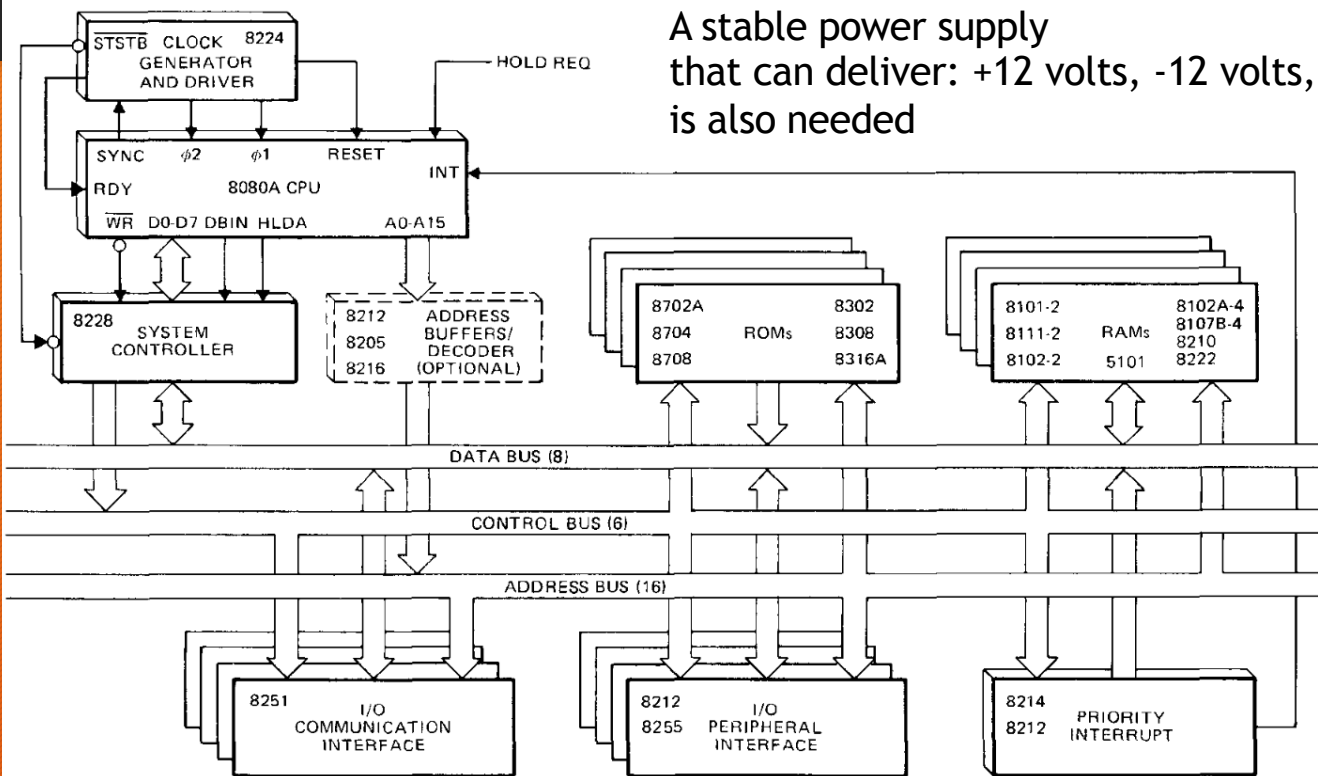
June 2025



# 8080 Microprocessor and Support Chips

13

A stable power supply  
that can deliver: +12 volts, -12 volts, and +5 volts  
is also needed



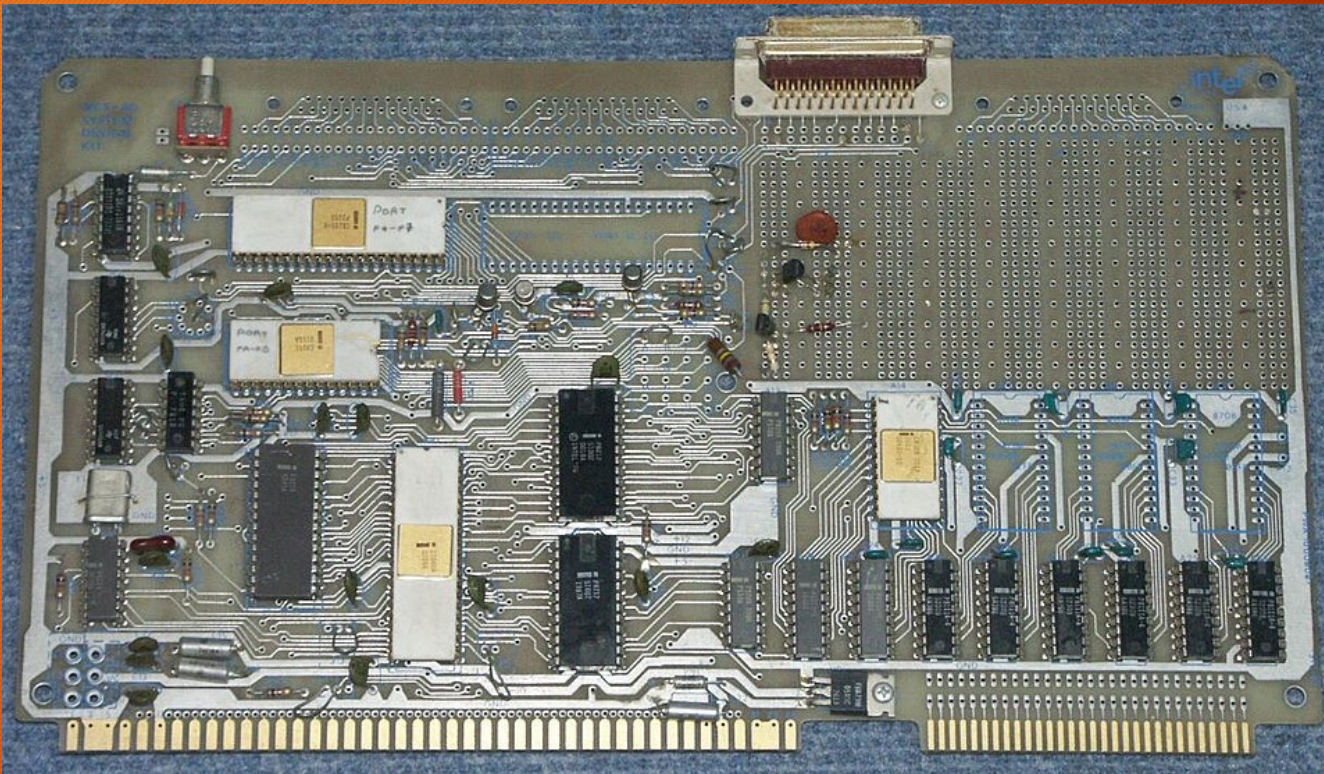
My first computer  
was an 8080 single board  
development kit with 256 bytes  
of RAM

It is now buried in  
the VCF Warehouse



# Early 8080 Microprocessor Development System

14



My first computer was an 8080 single board development kit with 256 bytes of RAM

It is now buried in the VCF Warehouse

June 2025



# Early Intel Microprocessor Development System

15



The Intellec 8 retailed for \$2,395 It was programmed in Assembly Language

June 2025



# Modern Microprocessors for the Hobbyist

16

- Today there are very powerful microprocessors that are available for very low cost (\$5-\$100 in the 2020s)
- Typical microprocessor used by Hobbyists (and some professionals) are:
  - Arduino Uno and its family members
  - Raspberry Pi
  - Adafruit Feather
  - Raspberry Pi Pico
  - Teensy
- There are others but they are less popular
- Languages used are C/C++, Python, Circuit Python



# Arduino Family

17

## Classic Family

- arduino uno R3
- arduino uno 4
- arduino uno 4
- Arduino UNO R4 Minima
- Arduino UNO R4 WiFi
- Arduino Leonardo
- Arduino UNO Mini Limited Edition
- Arduino Micro
- Arduino Zero
- Arduino UNO WiFi Rev2

## Mega Family

- Arduino Mega 2560 Rev3
- Arduino Due
- Arduino Giga R1 WiFi

## MKR Family

- Arduino MKR 1000 WiFi
- Arduino MKR WiFi 1000
- Arduino MKR Fox 1200
- Arduino MKR WAN 1300
- Arduino MKR WAN 1310
- Arduino MKR GSM 1400
- Arduino MKR NB 1500
- Arduino MKR Vidor 4000
- Arduino MKR Zero

## Nano Family

- nano
- Nano 33 IoT
- Nano 33 BLE Rev2
- Nano 33 BLE Sense Rev2
- Nano RP2040 Connect
- Nano Every
- Nano ESP32

June 2025



# Feather, Pico, and Teensy Microprocessors

18

## Feather Family

- Atmel (now Microchip) ATmega328P
- Atmel (Microchip) ATmega32u4
- Atmel (Microchip) ATSAMD21 Cortex M0
- Atmel (Microchip) ATSAMD51 Cortex M4
- Atmel (Microchip) ATSAME51 Cortex M4

## Pico

- Dual-core Arm Cortex-M0+ processor

## Teensy



# Digital Signal Processor (DSP)

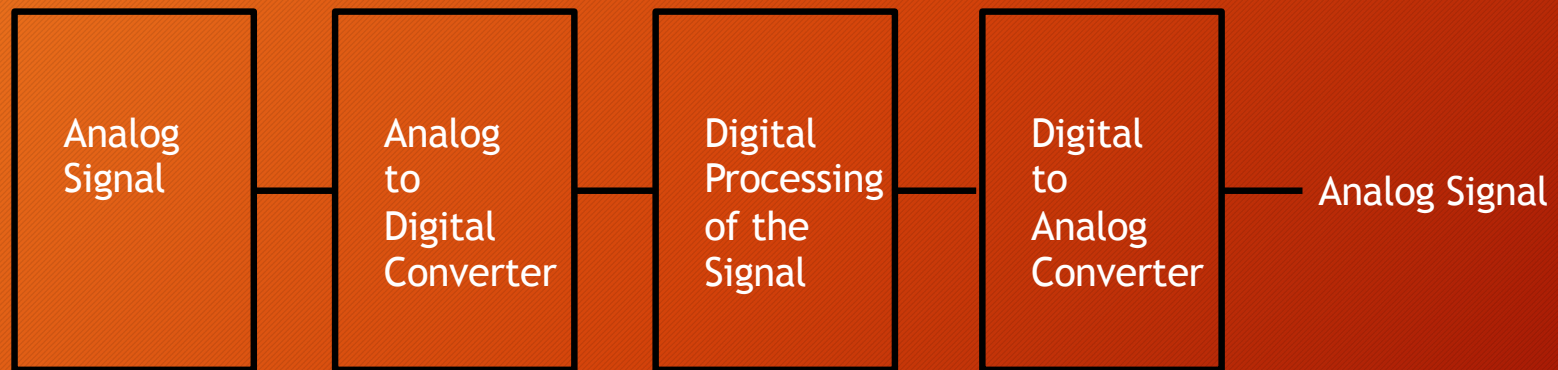
19

- A specialized microprocessor chip, with its architecture optimized for the operational needs of digital signal processing
- Processing of audio or radio signals requires complicated mathematical calculations; the DSP has instructions sets that perform these calculations significantly faster than a general purpose computer
- The operations on a DSP must be performed in real time
- There is no operating system
- Many DSPs only perform binary arithmetic
- They often have multiple cores to process data in parallel



# DSP Architecture

20





# How does a DSP Work?

21

- In analog circuits, all the processing of the signal is done by analog components: resistors, capacitors, inductors, semi-conductors
- We often analyze the circuit by examining how it works at different frequencies: frequency domain analysis
- In DSPs, all the processing of a signal is done by writing mathematical equations equivalent or similar to the analog circuits and solving the equations
- We analyze the code by examining how it works at different times: time domain analysis



# Differences in Programming a DSP compared to a non DSP

22

- All arithmetic is done in binary numbers
- Math libraries are too slow for real time processing
- For example calling a math library for sine, cosine or arctangent does not work
- Thus, a look up table is created for
  - sine and arctangent every 1 degree from 0-90 degrees
  - other angles are computed using symmetry in angles from 90-360 degrees
- Parallel processing using multiple core processors is necessary
- Coding is done in C/C++
- My approach was to develop/test the algorithms in MATLAB
- Then convert to Visual Studio C manually
- Better programmers would develop directly in C, but coding in C is complex compared to MATLAB



# Most DSPs have multiple cores

23

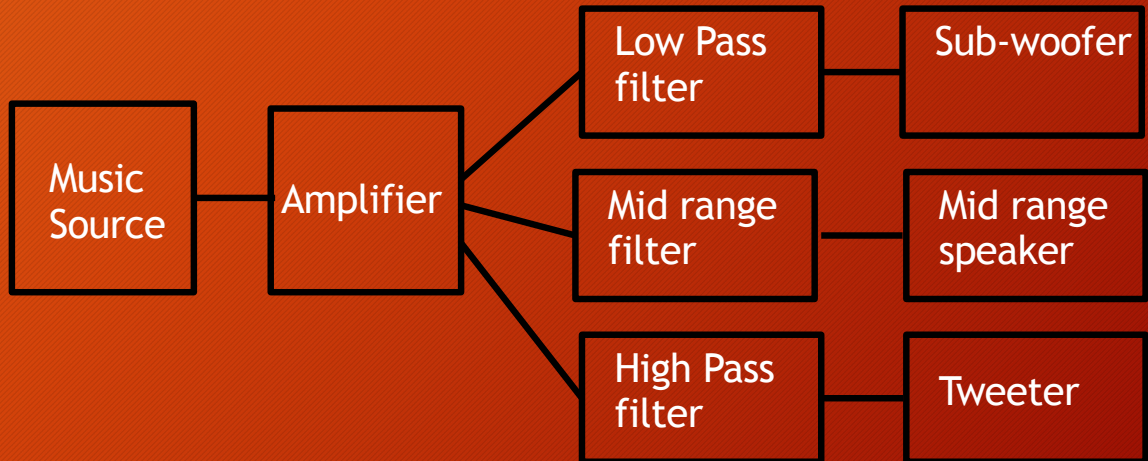
- Parallel processing is performed where ever possible
- So if the processor must perform an "if statement"
- Both paths of the if are computed in parallel,
- When the if evaluation is done, the unneeded path is abandoned and code continues on the correct path



# Example of Analog Processing

24

- Problem to be solved:
- We have an audio system and we want to
  - send low frequencies to the sub-woofer
  - Send higher frequencies to mid range speakers
  - Send the highest frequencies to the tweeter





# We examine the Problem in the frequency Domain

25

Low Pass  
filter



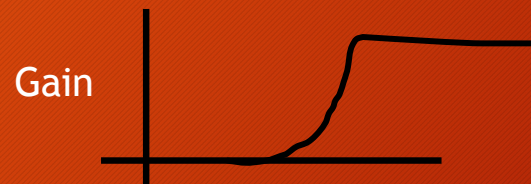
Filter\_LPF(f)

Mid range  
filter



Filter\_MRF(f)

High Pass  
filter



Filter\_HPF(f)

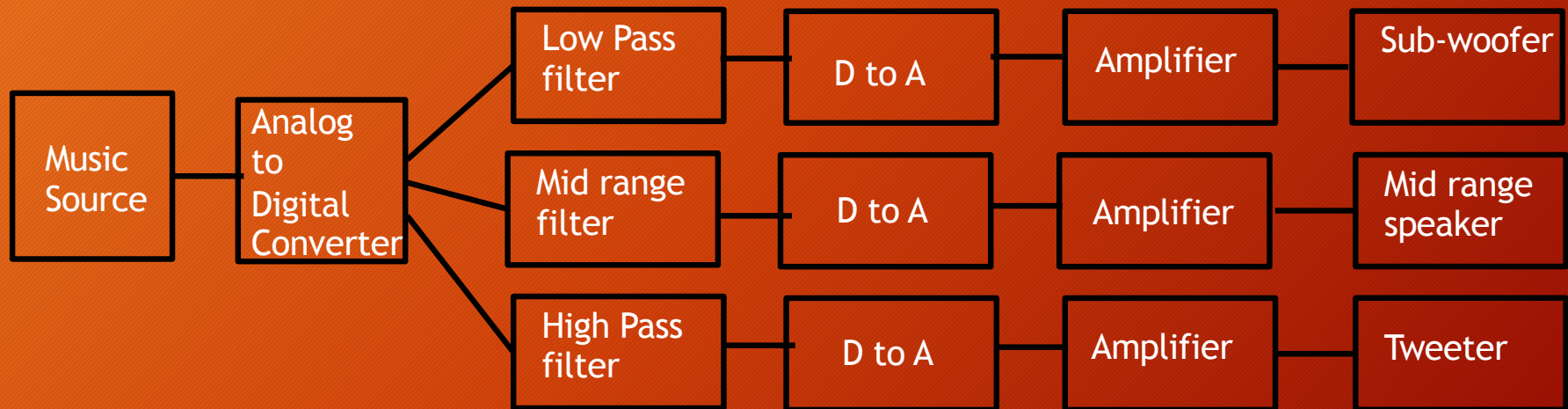
frequency

June 2025



# Example of Digital Processing

26



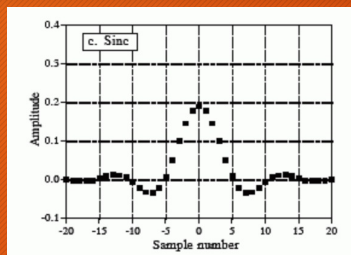


# We examine the Problem in the Time Domain

27

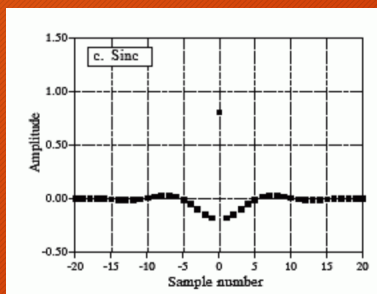
Filter\_LPF(t)

Low Pass  
filter



Mid range  
filter

High Pass  
filter



Filter\_HPF(t)

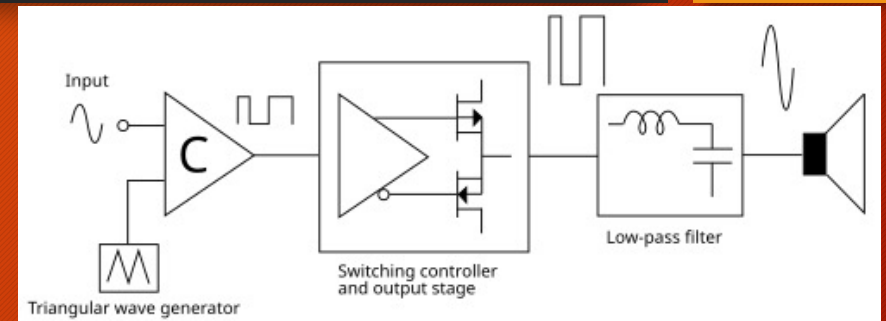
- We calculate the time function of the filter by taking the Fourier transform of the frequency function
- $filter(t) = \int_{-\infty}^{+\infty} filter(f)e^{i2\pi ft}df$
- We then sample the data
- Since these filters are very long, we need to shorten them
- We may have to slightly modify the resulting filter to avoid artifacts in the processing: this is called windowing



# Class D Audio Amplifier

28

- A class-D amplifier, uses mostly high power MOSFETs operating as switches, and not as linear gain devices as in analog amplifiers.
- The transistors rapidly switch back and forth between the supply rails, using pulse-width modulation, to produce a pulse train output.
- A simple low-pass filter is used to attenuate the high-frequency signal.
- Sometimes the loudspeaker is used as the filter
- Because the transistors are always either fully on or fully off, efficiency can exceed 90%.



[https://en.wikipedia.org/wiki/Class-D\\_amplifier](https://en.wikipedia.org/wiki/Class-D_amplifier)



<https://www.adafruit.com/product/5647>



# Questions

29